Introduction to Embryology

Objectives

- After this lecture you should be able to:
- Define Embryology.
- Define the developmental periods.
- Define the significance of embryology.
- Define the different embryological terminology.
- Define the nomenclature used to describe body parts, positions. and relationships.
- Describe in brief the important events in embryology

Definition and Significance of Embryology

Definition of Embryology:

- **Embryology** generally refers to the prenatal (قبل الولادة) development of embryos and fetuses.
- "**Human embryology**" is the science concerned with the origin and development of a human being from a zygote to birth of an infant.
- Development does not stop at birth. Important changes in addition to growth occur after birth (postnatal changes) e.g. development of teeth and female breasts.

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Importance of Embryology:

- The study of prenatal stages of development, especially those occurring during the embryonic period, help us **understand the normal body structure** and **the cause of congenital anomalies.**
- It is also concerned with studying various **genetic** and **environmental factors** that **disturb normal development and produce birth defects.**

➔ Developmental Periods

Prenatal development:

Includes the main developmental changes occuring before birth, **and is divided into 2 periods:**

Postnatal development:

Includes changes occurring after birth. e.g. teeth and breast.

Note:

Prenatal development is more rapid than postnatal development and results in more striking changes.

Note:

The most critical period is the **embryonic period.**

Embryonic period:

Begins at fertilization and ends with the end of the 8th week.

(called an embryo)

Fetal period:

Begins at the beginning of the 9th week and ends at birth.

(called a fetus)

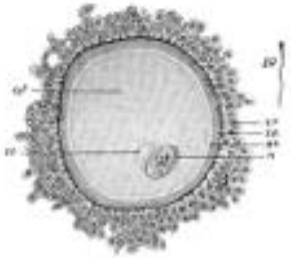
➔ Critical Periods of Human Development:

- It is the stage of development of an embryo that is susceptible to an agent, such as a drug or virus, which can lead to congenital abnormalities.
- The development of the embryo is most easily disrupted when the tissues and organs are forming during the embryonic period.

Common terms

- **Oocyte;** the immature ovum, female germ cell.
- **Ovum;** the mature female germ cell.
- **Sperm;** the mature male germ cell.
- **Zygote;** the fertilized ovum.
- **Cell division;** one cell divides into two cells; there are two types of cell division:
- **A- Mitotic;** the cell produces 2 cells each contains 44 autosomes and 2 sex chromosomes
- **B- Meiotic;** (reduction) it occurs in the primitive germ cells in the testes or the ovaries, it produces 2 cells each contains 22 autosomes and one sex chromosomes.

Cell division



oocyte

ovum

sperm

zygote

Descriptive terms

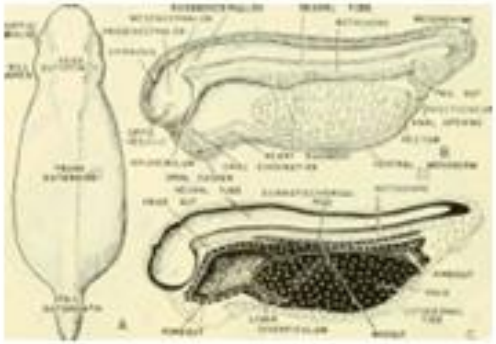
➔ Directions:

same

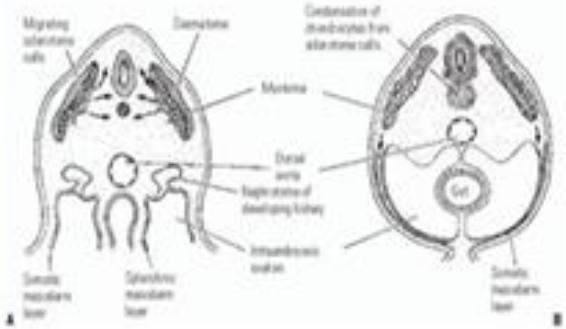
- Cranial; the **top** of the embryo or the head.
- Cephalic; **superior** or the head.
- Caudal; **inferior** or the tail end.
- Dorsal; **back** of the embryo.
- Ventral; **anterior** or the belly side.
- Medial; **near** to the midline.
- Lateral; **flank** side.

➔ Planes and sections:

- Longitudinal; **median** or **sagittal**. (cuts it to left & right)
- Coronal; **frontal**.
- Transverse; **horizontal.** (cuts it to superior & inferior)



Longitudinal



Transverse; horizontal

Major events during the embryonic period

- Gametogenesis → occurs at 1st week.
- Fertilization → occurs at 1st week.
- Implantation → begins one week after fertilization.
- Development of the Central Nervous System (CNS; brain & spinal cord) → occurs at 3rd week.
- Embryonic Folding → occurs at 4th week.

Note:
The Bilaminar & Trilaminar discs are supposed to be present before the embryonic folding.

Gametogenesis

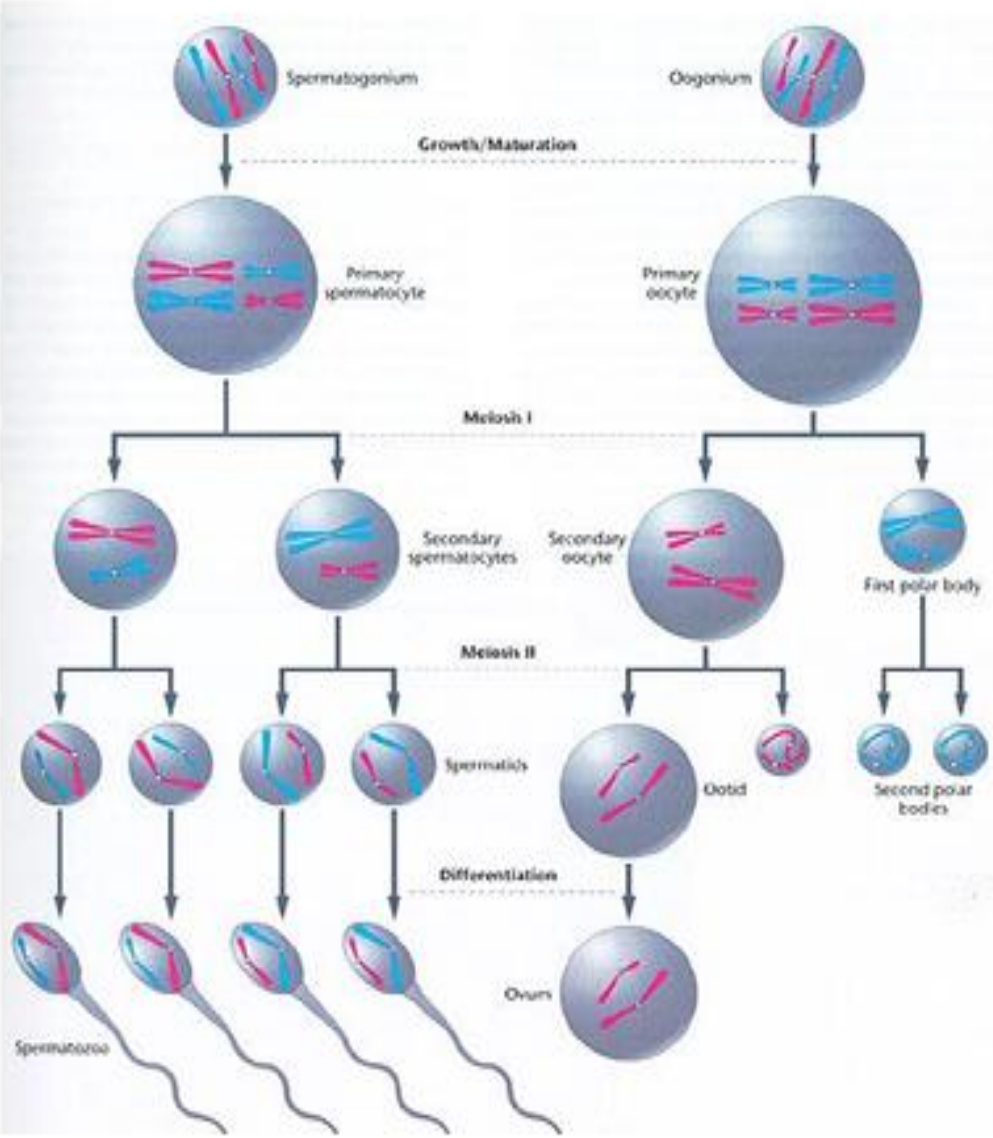
- It is the production of mature gametes (sperm and ovum) by gonads (testes in males and ovaries in females).

- Gametogenesis

1- Spermatogenesis 2- Oogenesis

Note:

- Spermatogonium and Oogonium are immature gametes (germ cells).
- **Meiosis** occurs in gametogenesis.



Gametogenesis in males and females	Spermatogenesis (Males)	Oogenesis (Females)
Definition	It is the process of formation of mature sperms	It is the process of formation of mature ovum
Site	takes place in the seminiferous tubules in the testis	occurs in the cortex of the ovary
Duration	occurs continuously from puberty till old ages	starts during fetal life, continues after puberty, and fertilization, till menopause
Result	1- Reduction of chromosomal number from the diploid to the haploid number. 2- Change the germ cell (spermatogonium) to the motile (mature) sperm. 3- Increase the number of the sperms.	It ends with a haploid (23 chromosomes) number of chromosomes

Fertilization:

- It is the process in which a male gamete (sperm) unites with a female gamete (ovum) to form a single cell (zygote).
- Site: It occurs in the uterine tube.

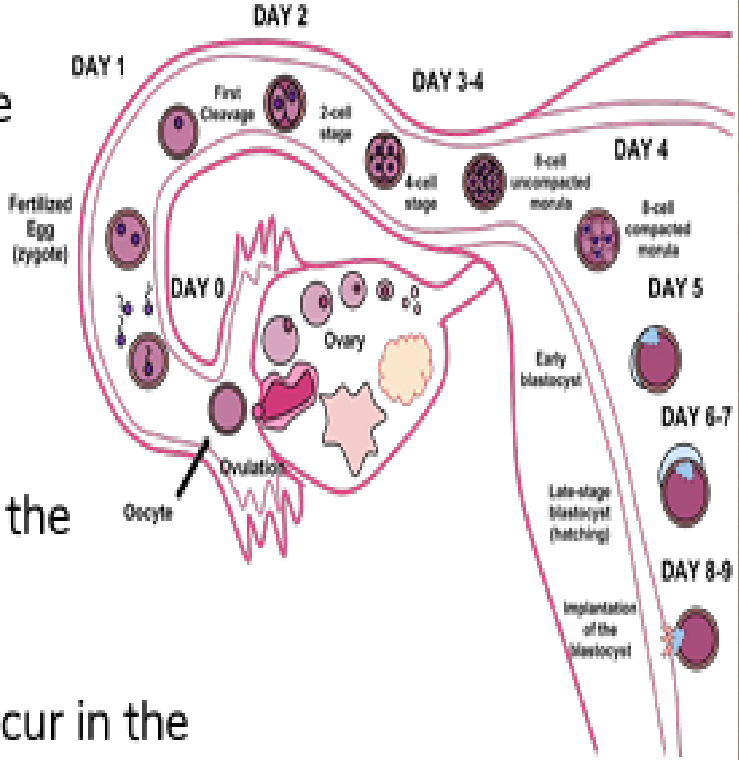
Implantation:

It is the process of embedding of the blastocyst in the endometrium of the uterus.

- It begins one week after fertilization.
- It is completed by the 12th day after fertilization.
- Sites of implantation:

  1- Normal site of implantation:
  In the upper part of the posterior surface of the uterus near the funds.

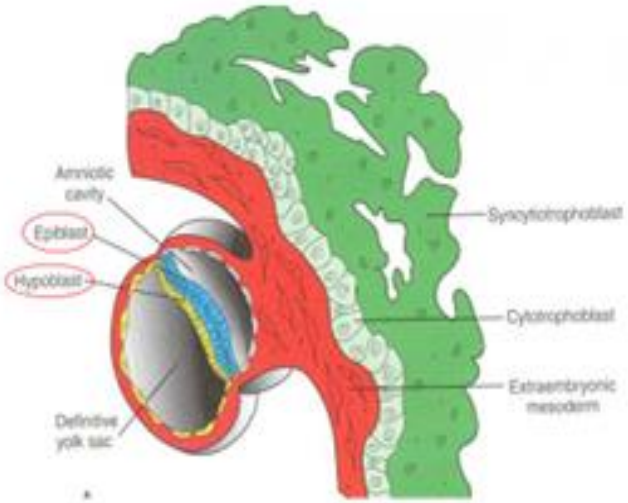
  2- Abnormal site of implantation (ectopic pregnancy): Most of ectopic pregnancies occur in the uterine tube.



Bilaminar disc:

It is the differentiation of the cells into two layers (in the 2nd week of pregnancy):

- Epiblast: High columnar cells adjacent to the amniotic cavity. (Dorsal)
- Hypoblast: Small cuboidal cells adjacent to Yolk sac. (Ventral)

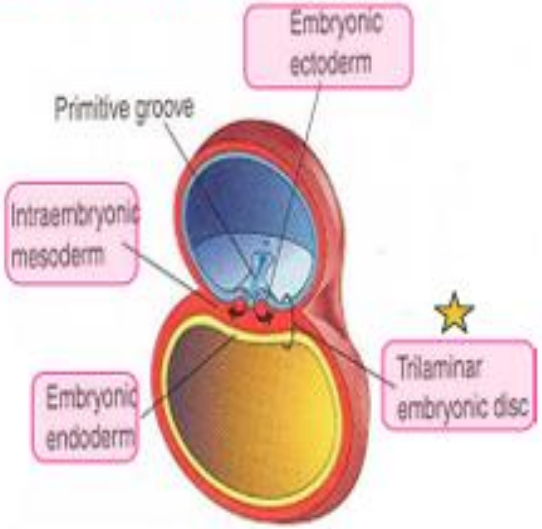


Trilaminar disc:

The embryonic disc is formed of 3 layers (in the 3rd week of pregnancy):

- Embryonic Ectoderm
- Intraembryonic Mesoderm.
- Embryonic Endoderm.

Cells in these layers will give rise to all tissues and organs of the embryo.



Gametogenesis

Spermatogenesis

- Production of sperm
- Results in four haploid sperm from each diploid cell that undergoes meiosis.

a. Undifferentiated germ cells called **spermatogonia** (diploid) undergo mitosis to produce daughter cells called **primary spermatocytes**.

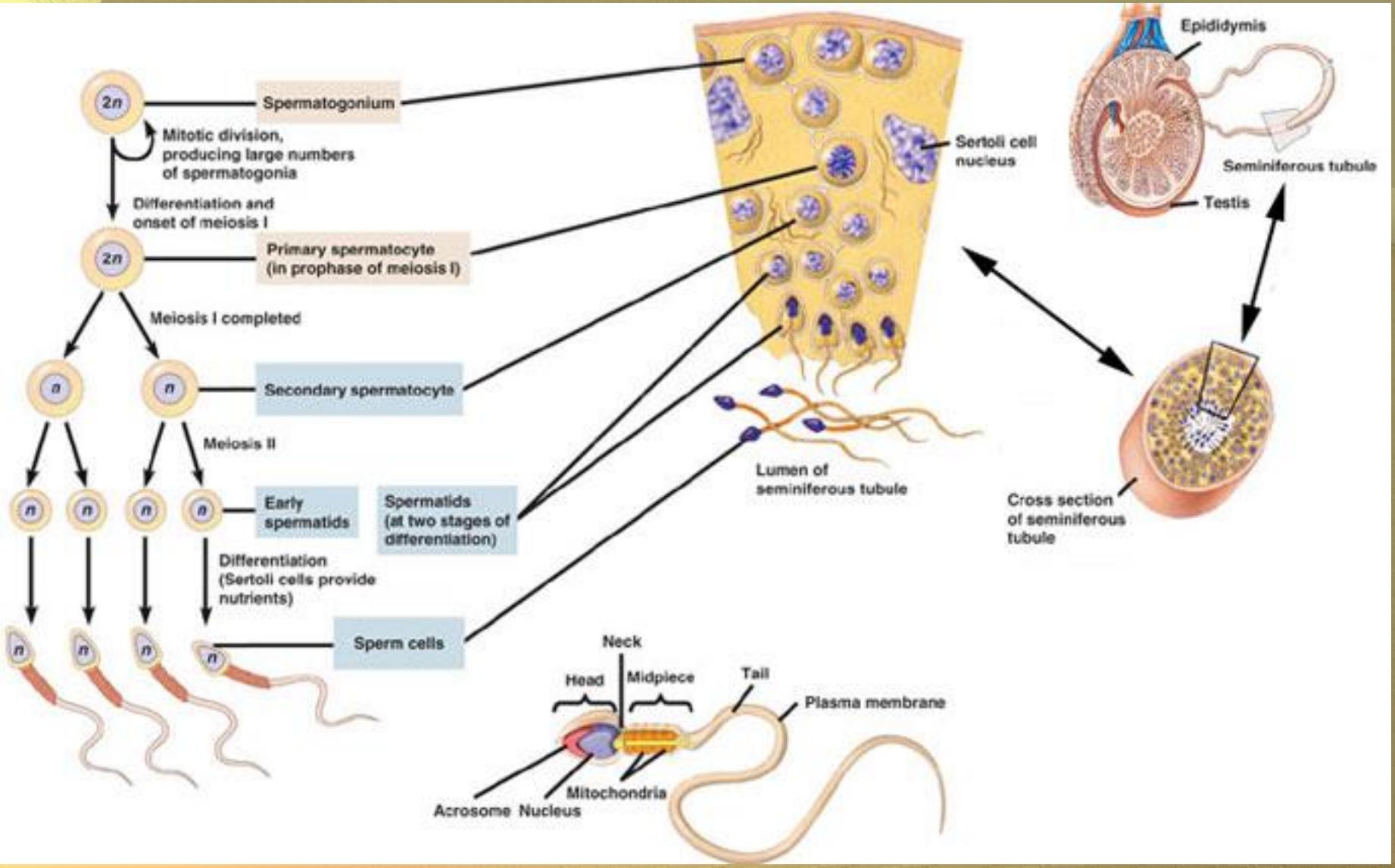
b. The primary spermatocytes undergo meiosis 1 to produce haploid **secondary spermatocytes** in a duplicated state.

c. The sister chromatids of each chromosome separate from each other during meiosis II. The daughter cells are haploid **spermatids**.

d. The spermatids will then eventually mature into functional sperm cells.

e. Sertoli cells provide the cells with nourishment and molecular signals.

Spermatogenesis



Oogenesis

- A normal baby girl had about 2 million primary oocytes in her ovaries.
- By 7 years old about 300,000 remain, her body reabsorbed the rest.

a. **Primary oocytes** have already entered meiosis I, but the nuclear division is arrested in a genetically programmed way.

- Meiosis will resume in one oocyte at a time, starting with the first menstrual cycle.
- Only about 400 to 500 oocytes will be released during her reproductive years.
- Follicle – primary oocyte and nourishing cell layers around it.

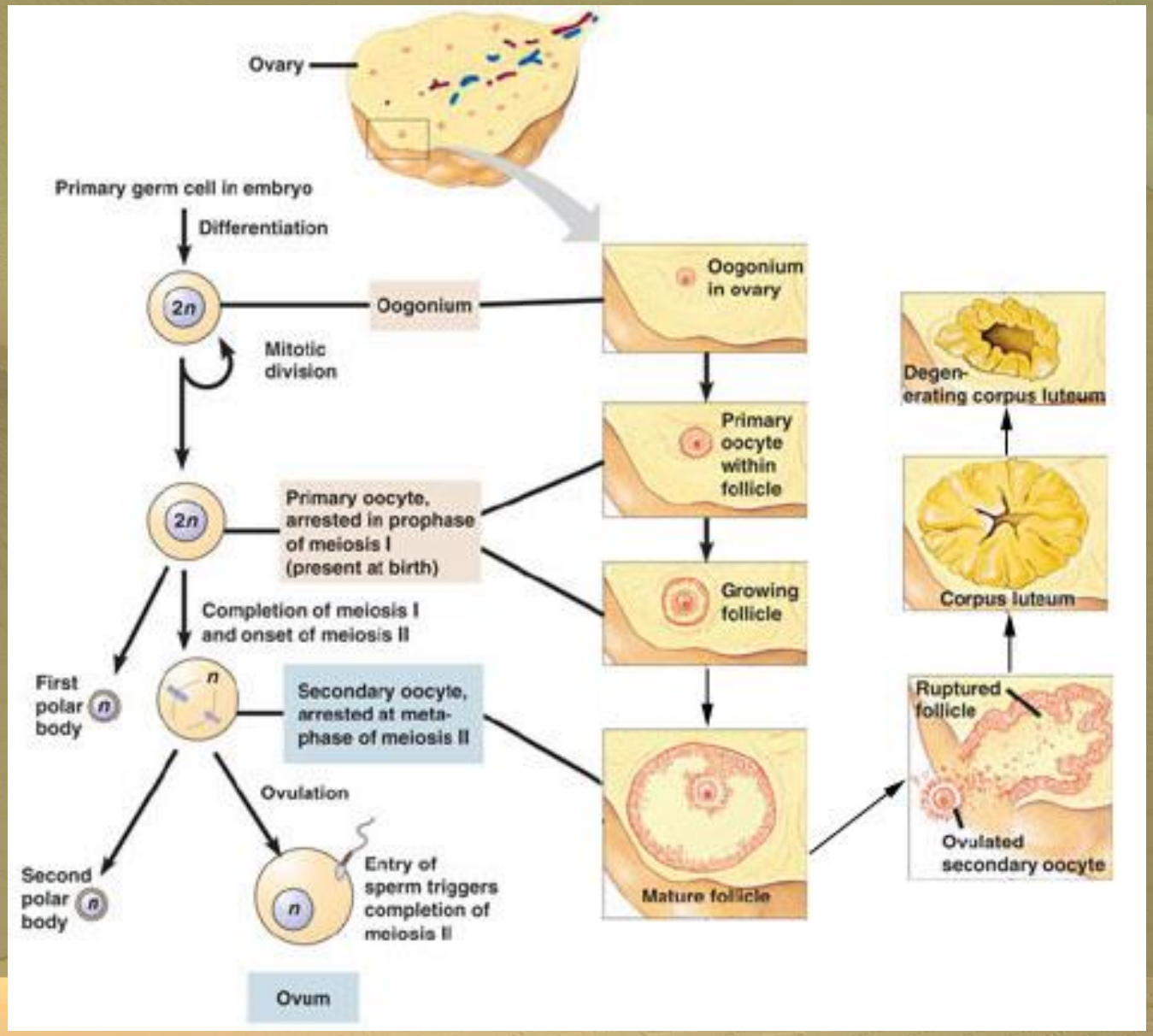
b. Stimulated by hormones the follicle continues to grow and the primary oocyte completes meiosis I. Resulting in the formation of a **secondary oocyte** (ends up with most of the cytoplasm) and the first of three polar bodies.

c. Ovulation then occurs releasing the secondary oocyte and the **polar body.**

d. Penetration of the sperm induces the secondary oocyte and the first polar body to complete meiosis II. There are now three polar bodies and one mature egg or ovum.

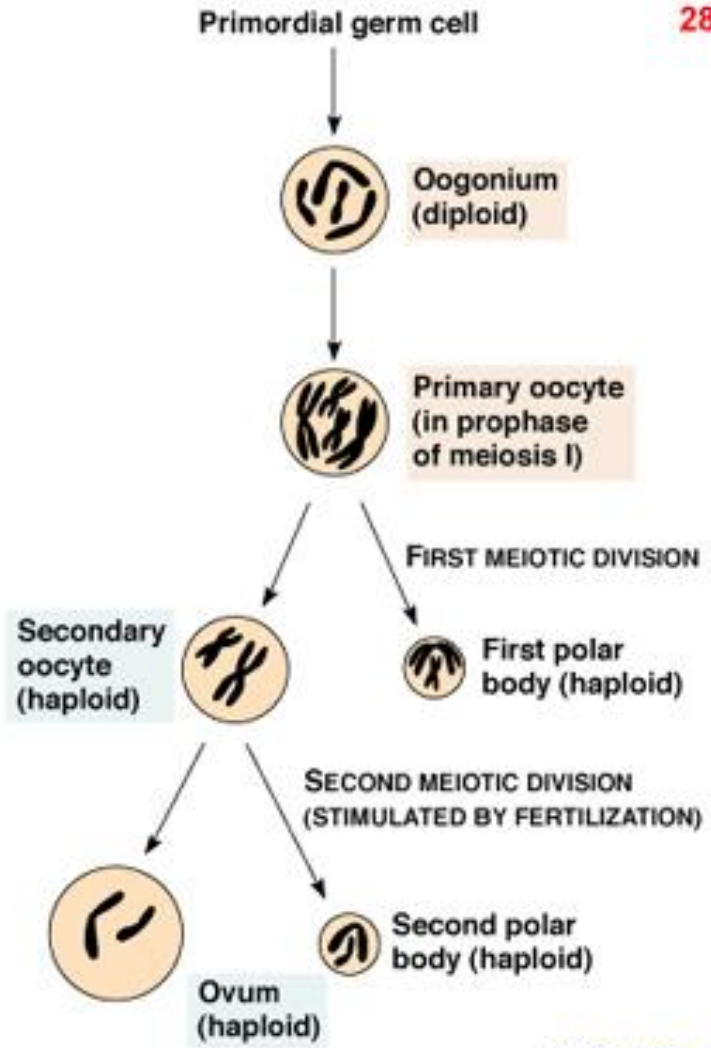
- As the sperm and egg nuclei fuse, their chromosomes restore the diploid number for a brand new zygote.

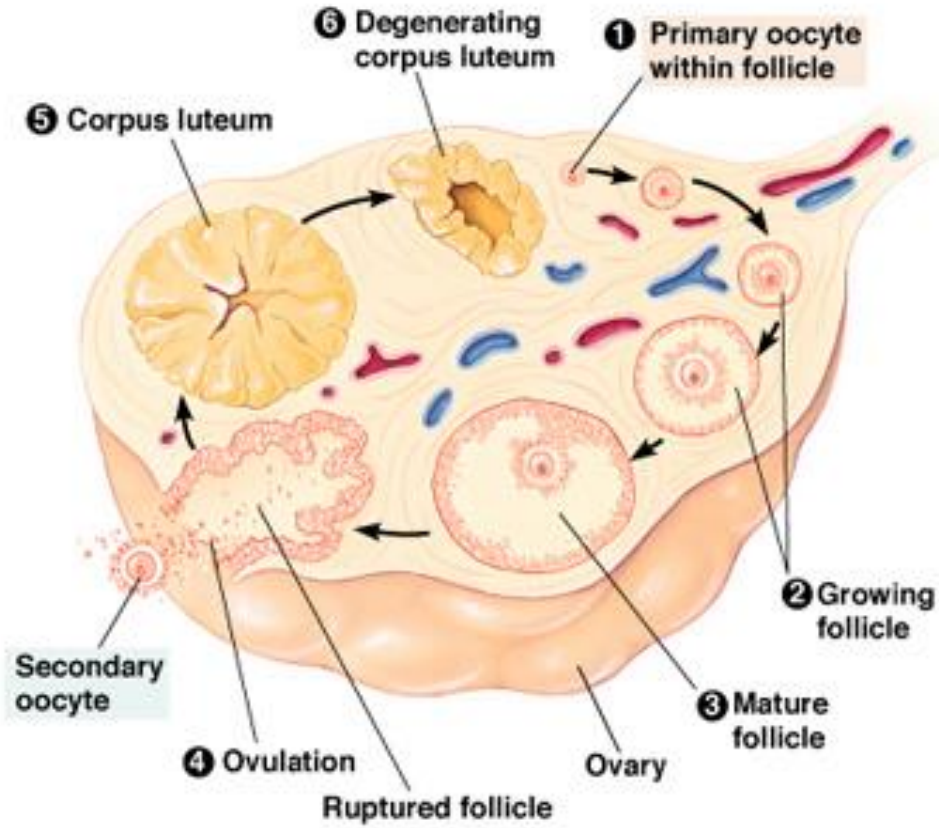
Oogenesis


Ovary
Primary germ cell in embryo
Differentiation
2n
Oogonium
Mitotic division
Oogonium in ovary
Primary oocyte within follicle
Growing follicle
Mature follicle
2n
Primary oocyte, arrested in prophase of meiosis I (present at birth)
Completion of meiosis I and onset of meiosis II
First polar body
n
n
Secondary oocyte, arrested at meta-phase of meiosis II
Ovulation
Second polar body
n
Entry of sperm triggers completion of meiosis II
Ovum
Ruptured follicle
Ovulated secondary oocyte
Corpus luteum
Degen-erating corpus luteum

Oogenesis

28d FSH --> stimulates follicle





LH stimulates ovulation

ovum to fertilization = GROWTH

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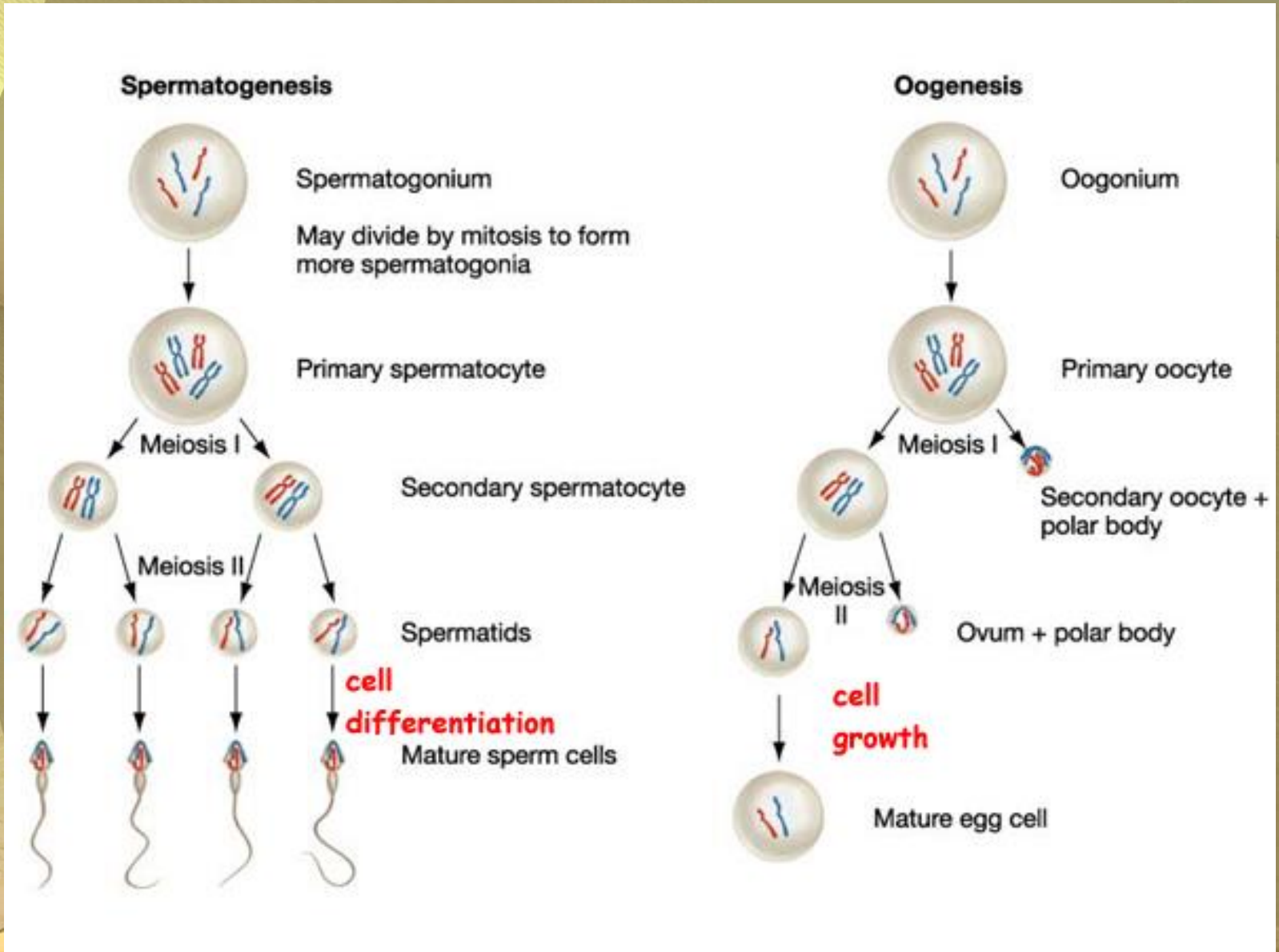
Comparing Spermatogenesis and Oogenesis



" لَقَدْ خَلَقْنَا الْإِنسَانَ فِي أَحْسَنِ تَقْوِيمٍ. "
(القرآن ٤:٩٥)